

A $2d - 1$ projection obstruction for three-point sorting embeddings

Strengthened partial packet for arXiv:2510.22186

Source and target

The source is Nadav Dym, Matthias Wellershoff, Efstratios Tsoukanis, Daniel Levy, and Radu Balan, *Quantitative Bounds for Sorting-Based Permutation-Invariant Embeddings*, arXiv:2510.22186 [1]. For $\mathbf{A} \in \mathbb{R}^{d \times D}$, the source studies

$$\beta_{\mathbf{A}}(\mathbf{X}) = (\text{sort}(\mathbf{X}a_1), \dots, \text{sort}(\mathbf{X}a_D)), \quad \mathbf{X} \in \mathbb{R}^{n \times d},$$

where sorting is columnwise and the rows of $\mathbf{X}$ are identified up to permutation. Orbit separation means that $\beta_{\mathbf{A}}(\mathbf{X}) = \beta_{\mathbf{A}}(\mathbf{Y})$ implies $\mathbf{X}$ and $\mathbf{Y}$ differ by a common row permutation.

The source proves an upper bound $D \geq n(d - 1) + 1$ for full-spark $\mathbf{A}$, and a logarithmic lower nonseparation bound. It then leaves open the sharpness of the bounds for $n > 2$.

The theorem is thus proven. $\square$

Remark 6. The two theorems above, Theorem 4 and Theorem 5 are stated in [MPv08] for the case $d = 2$ and using different but equivalent notation. Our contribution here is in extending the proof to the general case $d \geq 2$. Also, in [MPv08] it is shown that, for $d = 2$, the logarithmic lower bound is nearly attainable: there exist constants D_0 and c such that, for all generic matrices with $D \geq D_0$ rows, the mapping $\beta_{\mathbf{A}}$ is injective whenever $n \leq 2^{cD/\log D}$; or, equivalently, when $\log_2(n) \lesssim D/\log D$. It remains unclear whether similar bounds hold when $d > 2$.

We now visualize the preceding two results. According to Theorem 4 the map $\beta_{\mathbf{A}}$ separates orbits for full spark matrices $\mathbf{A} \in \mathbb{R}^{d \times D}$ with $d > 1$ whenever $D \geq n(d - 1) + 1$. In contrast, Theorem 5 shows that $\beta_{\mathbf{A}}$ fails to separate orbits (independently of the choice of $\mathbf{A}$) whenever $\lceil D/(d - 1) \rceil \leq \log_2(n) + 1$. For $n = 2$, these bounds coincide and yield the sharp threshold $D \geq 2d - 1$ for orbit separation (assuming $\mathbf{A}$ has full spark). Via the connection to real phase retrieval established in [BT23c], this recovers the classical result (see, e.g., [BCE06]) that $2d - 1$ measurements are necessary and sufficient for sign retrieval in $\mathbb{R}^d$ provided the measurement vectors form a full spark frame.

For $n > 2$, however, the upper and lower bounds no longer match. In general, it remains open whether either bound is sharp in this regime, although we present some tentative progress in this direction in Section 4. Figure 1

For $d = 2$, the source phase diagram marks unresolved gray-dot parameter pairs between the lower nonseparation curve and the upper full-spark line.

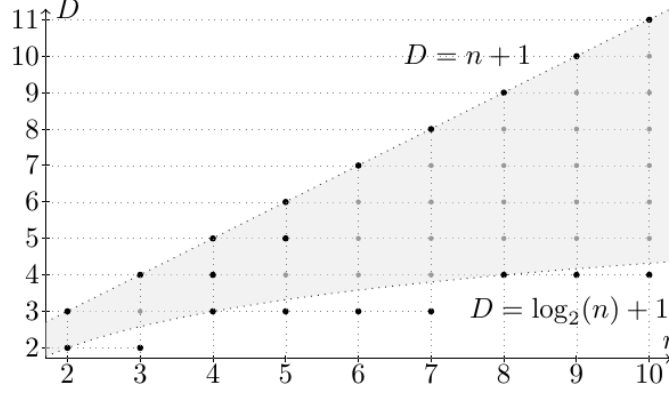


Figure 1: Phase diagram for orbit separation when $d = 2$. On and below the lower curve, $\beta_{\mathbf{A}}$ does not separate orbits for any choice of $\mathbf{A} \in \mathbb{R}^{2 \times D}$. On and above the upper line, $\beta_{\mathbf{A}}$ separates orbits provided that $\mathbf{A}$ has full spark. For larger d , the qualitative behavior remains the same, but the vertical scale changes by a factor of $(d - 1)$. Black dots indicate parameter pairs (n, D) for which it is known whether there exists a matrix $\mathbf{A}$ such that $\beta_{\mathbf{A}}$ separates orbits. Gray dots mark parameter pairs for which this question remains open. The two black dots at $(n, D) = (4, 4)$ and $(5, 5)$ correspond to cases where an explicit construction of such a matrix $\mathbf{A}$ is known (see Section 4).

Result

Theorem 1. *Let $d \geq 2$. If $D \leq 2d - 1$, then no matrix $\mathbf{A} \in \mathbb{R}^{d \times D}$ makes*

$$\beta_{\mathbf{A}} : \mathbb{R}^{3 \times d} / S_3 \longrightarrow \mathbb{R}^{3 \times D}$$

orbit separating. Equivalently, any sorting key that separates all three-point multisets in $\mathbb{R}^d$ must use at least $2d$ projections.

For $d = 2$, the source theorem gives orbit separation for every full-spark $\mathbf{A} \in \mathbb{R}^{2 \times D}$ once $D \geq 4$. Hence the exact planar threshold for three points is $D = 4$. For general d , the theorem strengthens the source lower bound in the $n = 3$ row from $D \leq 2d - 2$ to $D \leq 2d - 1$. In particular, it proves that the source’s numerically negative cases $(n, d, D) = (3, 3, 5)$ and $(3, 4, 7)$ fail for every choice of $\mathbf{A}$.

Proof intuition

After choosing d independent projection directions as coordinates, a $D = 2d - 1$ key has the form $[I_d \mid B]$, where B has $d - 1$ columns. Thus B^T has a nonzero kernel vector c . The construction uses this one missing linear constraint: choose different row permutations for different coordinate columns, but choose the coordinate vectors so that the change in every extra projection is the same kernel-weighted sum $\sum_i b_{ij} c_i w = 0$. Sorting cannot see the separate coordinate permutations. A single row permutation would have to realize two different 3-cycles at once, or fix one distinct coordinate while cycling another, which is impossible.

Proof

It suffices to prove the theorem for $D = 2d - 1$. If $D < 2d - 1$, append zero columns to $\mathbf{A}$; these add only zero output columns, so they cannot repair nonseparation.

First suppose $\text{rank } \mathbf{A} < d$. Choose a nonzero $z \in \mathbb{R}^d$ with $\mathbf{A}^T z = 0$. Let $\mathbf{X}$ have all rows equal to 0, and let $\mathbf{Y}$ have rows $z, 0, 0$. Then $\beta_{\mathbf{A}}(\mathbf{X}) = \beta_{\mathbf{A}}(\mathbf{Y})$, but $\mathbf{X}$ and $\mathbf{Y}$ are not in the same S_3 -orbit.

Now assume $\text{rank } \mathbf{A} = d$. Permuting the columns of $\mathbf{A}$ only permutes output columns of $\beta_{\mathbf{A}}$. Also, if $S \in GL(d, \mathbb{R})$, then $\beta_{S\mathbf{A}}(\mathbf{X}) = \beta_{\mathbf{A}}(\mathbf{X}S)$, and $\mathbf{X} \mapsto \mathbf{X}S$ is a bijection commuting with row permutations. Thus orbit separation is unchanged by these operations. We may therefore assume

$$\mathbf{A} = \begin{pmatrix} I_d & B \end{pmatrix}, \quad B \in \mathbb{R}^{d \times (d-1)}.$$

Choose a nonzero $c = (c_1, \dots, c_d)^T \in \ker B^T$, and let $J = \{i : c_i \neq 0\}$. Let C denote the 3-cycle

$$C(v_1, v_2, v_3)^T = (v_2, v_3, v_1)^T.$$

Assume first that $|J| \geq 2$. Pick distinct $p, q \in J$. Put

$$P_q = C^{-1}, \quad P_i = C \quad (i \in J, i \neq q), \quad P_i = I \quad (i \notin J).$$

Let $W = \{u \in \mathbb{R}^3 : u_1 + u_2 + u_3 = 0\}$. The maps $C - I$ and $C^{-1} - I$ restrict to invertible maps on W . Choose $w \in W$ outside the finite union of lines for which one of the vectors below has repeated coordinates, and for each $i \in J$ choose $x_i \in W$ satisfying

$$(P_i - I)x_i = c_i w.$$

For $i \notin J$, set $x_i = 0$. Let $\mathbf{X} \in \mathbb{R}^{3 \times d}$ have columns x_i , and let $\mathbf{Y}$ have columns $P_i x_i$.

For the first d coordinate projection columns of $\mathbf{A}$, the projected vectors are merely permuted, so their sorted values agree. For an extra column b_j of B ,

$$\mathbf{Y}b_j - \mathbf{X}b_j = \sum_{i=1}^d b_{ij}(P_i - I)x_i = \sum_{i \in J} b_{ij}c_i w = 0,$$

because $B^T c = 0$. Hence $\beta_{\mathbf{A}}(\mathbf{X}) = \beta_{\mathbf{A}}(\mathbf{Y})$.

The two matrices are not in the same row-permutation orbit. The entries of x_p and x_q are distinct by the choice of w . Any common row permutation R with $R\mathbf{X} = \mathbf{Y}$ would have to satisfy $Rx_p = P_p x_p$ and $Rx_q = P_q x_q$. Distinct entries force $R = P_p$ and $R = P_q$, but $P_p = C$ and $P_q = C^{-1}$.

It remains to handle $|J| = 1$, say $J = \{p\}$. Then the p -th row of B is zero. Choose $q \neq p$, set $P_p = C$ and $P_i = I$ for $i \neq p$, and choose $x_p, x_q \in \mathbb{R}^3$ with three distinct entries while setting all other $x_i = 0$. Define $\mathbf{X}$ and $\mathbf{Y}$ as before. The coordinate projections still agree after sorting. The extra columns of B do not see coordinate p , and all other coordinates are unchanged, so the extra projections agree as well. A common row permutation would have to be C from coordinate p , but would have to be I from coordinate q . This is impossible. The proof is complete.

Verification and limitations

The local run indexes were searched for the exact arXiv id, for sorting-based permutation-invariant embeddings, and for orbit-separation keywords; no existing packet for this strengthened $n = 3$ obstruction was found before this packet was written. A bounded literature check on 2026-07-18

searched exact phrases around $D = 2d - 1$, $n = 3$, $\beta_{\mathbf{A}}$, and sorting-based permutation-invariant embeddings. It found the source paper, the Balan-Haghani-Singh predecessor, and Matousek-Privetivy-Skovron’s projection paper [2], but no later exact answer to this $n = 3, D = 2d - 1$ obstruction.

The script `code/verify_three_projection_counterexample.py` checks the construction against the actual sorted-projection map for sampled normalized keys $[I_d \mid B]$ with $2 \leq d \leq 8$, including full-support and singleton-support kernel cases. The script is not a substitute for the proof; it is a sanity check that the displayed construction is implemented consistently.

This packet does not settle the source paper’s general dimension gap or any distortion question. It proves the sharp lower obstruction $D \geq 2d$ for three-point multisets. Existence at $D = 2d$ for every d is not proved here, although the source paper gives explicit positive examples for $(n, d, D) = (3, 3, 6)$ and $(3, 4, 8)$, and the $d = 2$ case follows from the source’s full-spark upper theorem.

Human-review recommendation

Likely valid strengthened partial result. Review should check the normalization $\mathbf{A} \mapsto S\mathbf{A}$, the linear-algebra kernel construction for $[I_d \mid B]$, and the row-permutation obstruction in the two support cases.

References

- [1] N. Dym, M. Wellershoff, E. Tsoukanis, D. Levy, and R. Balan, *Quantitative Bounds for Sorting-Based Permutation-Invariant Embeddings*, arXiv:2510.22186; IEEE Trans. Inf. Theory 72 (2026), no. 6, 4297–4311.
- [2] J. Matousek, A. Privetivy, and P. Skovron, *How many points can be reconstructed from k projections?*, SIAM J. Discrete Math. 22 (2008), no. 4, 1605–1623.